

Marginal zone lymphoma

The **antigen-driven lymphoma** — extranodal (MALT), splenic and nodal MZL share a CD5–/CD10– phenotype but not a treatment. **Find and treat the driver first** (H. pylori, HCV, C. psittaci), irradiate localized disease with curative intent, and reserve systemic therapy for disseminated or symptomatic disease

Compiled from primary trial reports and consensus guidelines (NCCN; WHO-HAEM5; EGILS) verified against PubMed, September 2026.

THREE (NOW FOUR) ENTITIES — WHO-HAEM5

Extranodal MZL of MALT <i>most common</i>	Arises in mucosa made lymphoid by chronic antigen. Stomach most common ; then ocular adnexa, lung, salivary gland, thyroid, skin, breast, dura, small bowel. Lymphoepithelial lesions. Usually localized — often curable locally
Splenic MZL	Spleen + marrow (intrasinusoidal) ± blood; splenomegaly, cytopenias, villous lymphocytes , small IgM paraprotein, autoimmune cytopenias. Nodes uncommon. HCV -linked subset. del(7q), NOTCH2, KLF2 mutations
Nodal MZL	Nodes only (no extranodal or splenic primary). Behaves and is treated like FL
Primary cutaneous MZL	Split off as its own entity in WHO-HAEM5 ; very indolent, skin-directed therapy. Borrelia link in European series

All are CD5– CD10– small B cells, but genetics differ by site: t(11;18)(q21;q21) **BIRC3::MALT1** in gastric and lung MALT; **TNFAIP3** loss in ocular adnexa; **GPR34** in salivary gland; **FAS** in cutaneous MZL; **KMT2D, NOTCH2, KLF2** in nodal MZL. Also t(14;18)(q32;q21) **IGH::MALT1**, t(1;14) **IGH::BCL10**, t(3;14) **IGH::FOXp1**.

WORKUP

- **Every MZL:** **HCV antibody + RNA, HBsAg + anti-HBc, HIV**, SPEP/immunofixation (paraprotein common), LDH, β2M
- **Gastric:** EGD with multiple biopsies (**map the stomach**), *H. pylori* by histology + stool antigen or breath test (serology if on PPI), **EUS** for depth and perigastric nodes, **FISH for t(11;18)**
- **Splenic:** smear (villous lymphocytes), flow, marrow biopsy, CT; DAT and autoimmune screen
- **PET/CT** is less reliable than in FL (MALT and SMZL are variably FDG-avid) — CT is acceptable; **a hot focus raises concern for transformation**
- Autoimmune history (Sjögren, Hashimoto) and site-specific exams: ophthalmology, ENT, thyroid ultrasound

ANTIGEN DRIVER → ANTIMICROBIAL FIRST

Site	Driver	Antimicrobial therapy & evidence
Stomach	<i>H. pylori</i>	Eradication is first therapy for localized disease. Remission 77.5% (32 studies, 1,408 pts): stage I 78% vs II1 56%; mucosa/submucosa 82% vs deeper 54%; t(11;18)+ only 22% vs 78%
Ocular adnexa	<i>Chlamydia psittaci</i> (regional)	Doxycycline 3 wk: ORR 65%, 5-yr PFS 55% (phase 2, stage I); better if Cp eradicated (86% vs 47%). Cp DNA in 89% of that Italian cohort — prevalence varies widely by region
Skin	<i>Borrelia burgdorferi</i> (Europe)	Antibiotic trial in Borrelia-positive cases
Small bowel (IPSID)	<i>Campylobacter jejuni</i>	α-heavy-chain disease ; early disease regresses with antibiotics
Spleen	Hepatitis C	Antiviral therapy first if no urgent indication: 7/9 CR after interferon with HCV clearance, none in HCV– (Hermine, NEJM 2002); DAAs — SVR 98%, lymphoma response 73% in MZL (Arcaini, Blood 2016)
Parotid · thyroid	Sjögren · Hashimoto	Autoimmune drive — no antimicrobial; treat by stage

STAGING GASTRIC MALT — LUGANO (GI) SYSTEM

I	Confined to the GI tract (single or multiple non-contiguous lesions)
II1 / II2	Into the abdomen: local (perigastric) nodes / distant (mesenteric, para-aortic) nodes
IIIE	Penetrates serosa into adjacent organs or tissues
IV	Disseminated extranodal, or supradiaphragmatic nodes. <i>No stage III</i>

Non-gastric and nodal/splenic disease: Ann Arbor/Lugano.

SMALL B-CELL DIFFERENTIAL

	CD5	CD10	CD23	Clue
MZL	–	–	–	Diagnosis of exclusion; cyclin D1– , MYD88 usually wild-type
CLL/SLL	+	–	+	CD200+, dim slg / CD20; LEF1+
Mantle cell	+	–	–	Cyclin D1, SOX11 , t(11;14)
Follicular	–	+	–/+	BCL2+ follicles, t(14;18) IGH::BCL2
LPL / Waldenström	–	–	–	MYD88 L265P (>90%), IgM, plasmacytic
Hairy cell	–	–	–	CD11c, CD25, CD103, CD123, annexin A1; BRAF V600E

Splenic MZL vs hairy cell: SMZL villi are short and **polar**, CD25–/annexin–/BRAF wild-type, and the marrow is intrasinusoidal and nodular; HCL has circumferential projections, a “dry tap”, monocytopenia. **Splenic B-cell lymphoma/leukemia with prominent nucleoli** (WHO-HAEM5; absorbs HCL-variant) is CD25–, annexin–, TRAP–, CD123–. Splenic diffuse red pulp lymphoma is the other mimic.

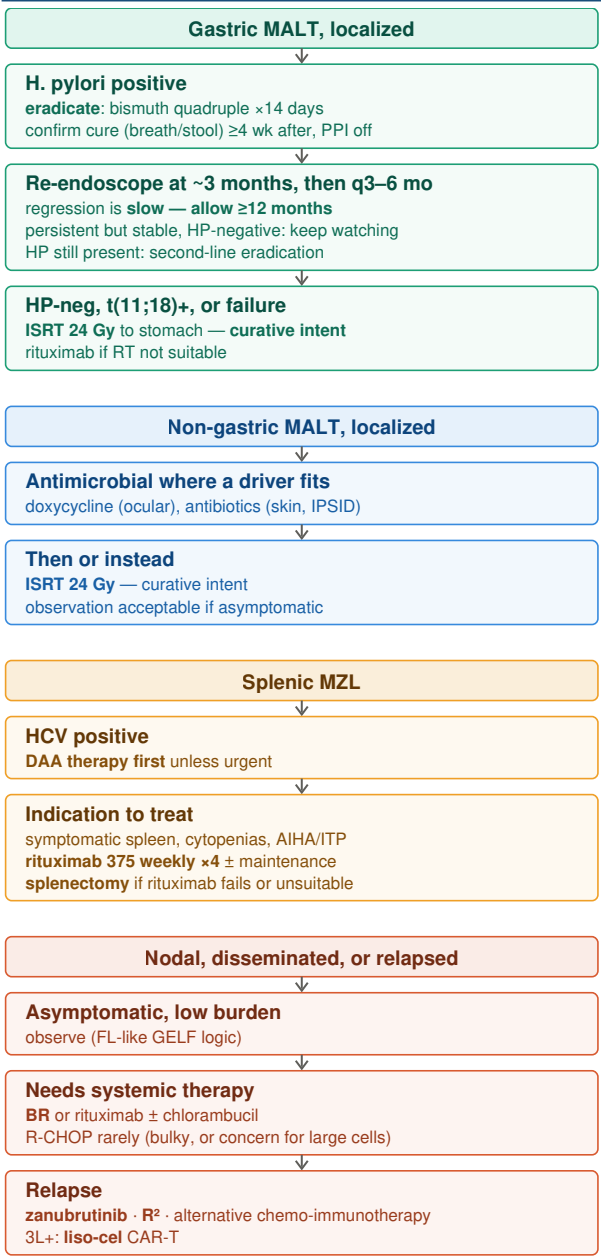
MALT-IPI — 1 POINT EACH

Risk	Points	5-yr EFS
Low	0	70%
Intermediate	1	56%
High	≥2	29%

Age ≥70 · stage III–IV · LDH >ULN. Derived in IELSG-19 (n=401), validated in 633; prognostic in gastric and non-gastric disease (Thieblemont, Blood 2017).

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TREATMENT PATHWAY



SYSTEMIC THERAPY — EVIDENCE AND STATUS

Agent · US status	Evidence
Rituximab ± chlorambucil	IELSG-19 (MALT, n=454): 5-yr EFS 68% R-chlorambucil vs 50% rituximab vs 51% chlorambucil; PFS better; OS ~90% in all arms
Bendamustine-rituximab	Phase 2 MALT data (Salar, Blood 2017) and StiL NHL1 (which included MZL) support BR as the usual frontline chemo-immunotherapy for advanced disease
Obinutuzumab <i>not approved for MZL</i>	GALLIUM MZL subgroup (n=195): no PFS benefit (4-yr 72.6% vs 64.1%) and more grade 5 AEs (14.9% vs 9.7%) — use rituximab
Zanubrutinib <i>accelerated, 2L+ after anti-CD20 (Sep 2021)</i>	MAGNOLIA (single arm, n=68): ORR 68%, CR 26%; 24-mo PFS 71%. AF 3%. Confirmatory MAHOGANY (zanubrutinib + rituximab vs R ²) still ongoing; approval remains accelerated
Ibrutinib <i>MZL indication withdrawn Dec 2023</i>	PCYC-1121 ORR 48% led to 2017 accelerated approval; SELENE (FL/MZL, + chemo) missed PFS → withdrawn
R ² <i>regular approval FL + MZL (2019)</i>	AUGMENT included 63 MZL; overall PFS HR 0.46 — but the MZL subgroup showed no significant PFS benefit (small, imbalanced). Approved nonetheless
Liso-cel <i>approved 3L+ (Dec 2025) — first CAR-T for MZL</i>	TRANSCEND FL MZL cohort (single arm; nodal 48%, splenic 27%, MALT 25%): ORR 95% of 66 infused (84% of all leukapheresed, per label), CR 56% ITT; grade 3 CRS / neuro 4% each
Axi-cel <i>not approved for MZL</i>	ZUMA-5 MZL cohort: ORR 77%, median PFS not reached at 3 yr
Umbralisib, other PI3Ki <i>withdrawn</i>	Umbralisib MZL/FL approval withdrawn 2022 (excess deaths). No bispecific is approved for MZL
Transformation to DLBCL is uncommon but prognostically important: suspect with rising LDH, a rapidly growing or FDG-hot lesion, B symptoms. Biopsy, then treat as DLBCL . Scattered large cells in a MALT lymphoma are not by themselves transformation — sheets are.	

KEY EVIDENCE

Zullo meta-analysis	H. pylori eradication: 77.5% remission; t(11;18)+ 22%
Ferreri phase 2	Doxycycline, ocular adnexal MZL: ORR 65%
Hermine / Arcaini	HCV clearance regresses splenic MZL (IFN; DAAs 73%)
IELSG-19	R-chlorambucil > either alone for EFS; OS equal
GALLIUM (MZL)	Obinutuzumab: no PFS gain, more fatal AEs
MAGNOLIA	Zanubrutinib R/R: ORR 68%, single arm
AUGMENT	R ² : benefit driven by FL, not MZL
TRANSCEND FL (MZL)	Liso-cel 3L+: ORR 95% infused, CR ~56%

AFTER ERADICATION — RESTAGING ENDOSCOPY

H. pylori	Lymphoma	Action
Gone	Gone	Observe: EGD q3–6 mo, then yearly
Gone	Persists	Asymptomatic and stable: keep observing (≥12 mo). Symptomatic or progressing: ISRT
Present	Gone	Second-line eradication
Present	Persists	Second-line eradication; ISRT if progressing or symptomatic

SPLENIC MZL — PRACTICAL POINTS

- **Many never need treatment** — observe asymptomatic, stable disease
- **Rituximab alone has largely replaced splenectomy** as first therapy: high response rates, no surgical risk. Splenectomy remains diagnostic and therapeutic when tissue is needed or rituximab fails
- **Before splenectomy:** pneumococcal, meningococcal (ACWY + B) and Hib vaccines ≥2 weeks before — **and before rituximab**, which blunts vaccine response
- Treat **autoimmune cytopenias** as part of the disease — they often respond to rituximab

DOSING

Regimen	Dosing
H. pylori bismuth quadruple	PPI BID + bismuth subsalicylate 524 mg QID + tetracycline 500 mg QID + metronidazole 500 mg TID–QID, 14 days
Rituximab	375 mg/m² weekly ×4 ; maintenance 375 q8wk ×2 yr optional
BR	Bendamustine 90 mg/m ² d1+2 + rituximab 375 d1, q28d ×6
R-chlorambucil <i>IELSG-19</i>	Chlorambucil 6 mg/m ² /d PO wk 1–6, then 2-wk blocks (wk 9–10, 13–14, 17–18, 21–22) + rituximab 375 wk 1–4, 9, 13, 17, 21
Zanubrutinib	160 mg BID or 320 mg daily, continuous
R ²	Lenalidomide 20 mg d1–21 q28d ×12 + rituximab weekly C1, d1 C2–5
ISRT	24 Gy / 12 fx ; 2 × 2 Gy for palliation

PEARLS

- **Always test for the driver** — H. pylori, HCV — before any cytotoxic therapy
- **t(11;18) predicts H. pylori non-response** (and dissemination): go to RT rather than repeated antibiotics
- **Give gastric MALT time:** histologic regression after eradication can take a year or more
- **MYD88 L265P favors LPL**; CD5+ or cyclin D1+ is not MZL